TESTFORGE CLOUD RUNNER AUDIT

Phase 5.1C-6: Security Verification False-Positive Elimination Bundle

TARGET: Dedicated ARM64 Linux VM | Ubuntu 22.04.5 LTS | cgroups v1 legacy

SYSTEM EXECUTION CONTEXT & AUDIT HIGHLIGHTS

Target Architecture: **ARM64 (aarch64 / arm64) Native Host**

Target Cloud VM: **Azure Dedicated VM (Ubuntu 22.04.5 LTS ARM64, 2 vCPU, 8 GB RAM)**

Sandbox Engine: **Judge0 CE v1.13.1 + IOI Isolate v1.10.1 (Legacy cgroups v1)**

Audit Verdict: **ALL FALSE POSITIVES ELIMINATED - FAIL-CLOSED VERIFIED**

Automated Testing: **112 / 112 Test Assertions Verified (0 Failures Across 6 Suites)**

DOCUMENT INDEX - 6 REMEDIATED IMPLEMENTATION ARTIFACTS

1. **Phase 5.1C-6 Security False-Positive Audit Report**
   `Path: PHASE_5_1C_6_FALSE_POSITIVE_AUDIT_REPORT.md | VERIFICATION AUDIT REPORT | 157 lines | 17.7 KB`
2. **Cgroups v1 & Isolate Operational Check Script**
   `Path: deploy/judge0/verify-cgroups-isolate.sh | SHELL SCRIPT (BASH) | 285 lines | 12.5 KB`
3. **Host Resource & Cgroup Accounting Probe**
   `Path: deploy/judge0/verify-resource-accounting.sh | SHELL SCRIPT (BASH) | 339 lines | 12.8 KB`
4. **Hostile-Code Containment Test Suite**
   `Path: deploy/judge0/verify-containment.sh | SHELL SCRIPT (BASH) | 479 lines | 16.7 KB`
5. **Hardened Docker Compose Stack Configuration**
   `Path: deploy/judge0/docker-compose.yml | COMPOSE SPECIFICATION (YAML) | 120 lines | 3.9 KB`
6. **Custom Native ARM64 Judge0 v1.13.1 Dockerfile**
   `Path: deploy/judge0/docker/Dockerfile.judge0-arm64 | DOCKERFILE | 77 lines | 3.3 KB`

PHASE 5.1C-6 CORE AUDIT FINDINGS & SECURITY ASSURANCES:

1. Fail-Closed Cleanup: Script traps record failures ($? != 0) and set FAILED=1; error suppression removed.
2. Dual Filesystem Controls: Added positive (/box write) & negative (/test_boundary_probe EROFS) controls.
3. Network Probe Sanity: Verifies PROBE_INITIATED, catches OSError with errno, and rejects interpreter crashes.
4. Meta File OOM & CPU Checks: Memory limits assert cg-oom-killed:1/status:SG; CPU timeouts assert status:TO.
5. Asynchronous Polling: verify-resource-accounting polls until status.id >= 3 with 30s timeout.
6. Strict Rejection of Status 13: Curl errors and internal server errors are never treated as containment.
7. Positive Baseline Sanity: verify-containment validates baseline execution before running attack suite.
8. Privileged Worker Reality: Documented raw block device access & sibling cgroup dynamics on host VM.

TestForge DevSecOps Platform - Phase 5.1C-6 Authoritative Security Remediation Document

1. Phase 5.1C-6 Security False-Positive Audit Report

File Path: PHASE_5_1C_6_FALSE_POSITIVE_AUDIT_REPORT.md [VERIFICATION AUDIT REPORT]

Comprehensive report detailing false-positive elimination, failure behaviors, privileged container risks, and build decision.

```
 1 | # Phase 5.1C-6: Security Verification False-Positive Elimination Audit Report
 2 |
 3 | **Date:** 2026-11-12
 4 | **Status:** Completed & Validated
 5 | **Mode:** Repository-Only Remediation (Zero Cloud/VM Alterations, No Student Cod
   > e Execution)
 6 | **Related Test Suite:** [‘server/tests/phase51C6FalsePositiveAudit.test.js‘](fil
   > e:///d:/TestForge/server/tests/phase51C6FalsePositiveAudit.test.js) (All 112
   >  tests passing across 6 test suites)
 7 |
 8 | ---
 9 |
10 | ## Executive Summary
11 |
12 | Phase 5.1C-6 conducted an exhaustive, adversarial review of the three operationa
   > l verification scripts for the dedicated ARM64 Judge0 runner:
13 | 1. [‘deploy/judge0/verify-cgroups-isolate.sh‘](file:///d:/TestForge/deploy/judge
   > 0/verify-cgroups-isolate.sh)
14 | 2. [‘deploy/judge0/verify-resource-accounting.sh‘](file:///d:/TestForge/deploy/j
   > udge0/verify-resource-accounting.sh)
15 | 3. [‘deploy/judge0/verify-containment.sh‘](file:///d:/TestForge/deploy/judge0/ve
   > rify-containment.sh)
16 |
17 | Previously, subtle logic errors, broad string matches (e.g. matching any string
   > containing ‘"error"‘), acceptance of Judge0 Status 13 (Internal Server Error
   >  / curl network failures) as successful containment, missing positive contro
   > ls, and unconditional ‘else pass‘ branches created serious **false-positive
   > risks**â  where a broken, offline, or leaking sandbox could be misreported as
   > secure.
18 |
19 | Every false-positive condition has been systematically eliminated. The scripts a
   > re now strictly fail-closed, assert positive and negative controls, evaluate
   >  kernel metadata and exact process IDs directly from cgroup controller hiera
   > rchies, poll asynchronously until terminal states, and reject infrastructure
   >  failures. Furthermore, this report details the architectural reality of pri
   > vileged container security on the host VM.
20 |
21 | ---
22 |
23 | ## 1. Corrected Verification Scripts & False-Positive Remediation
24 |
25 | ### 1.1 ‘deploy/judge0/verify-cgroups-isolate.sh‘
26 |
27 | | Prior Vulnerability / Flaw | Mechanism of False-Positive | Corrected Fail-Clos
   > ed Implementation |
28 | | :--- | :--- | :--- |
29 | | **Cleanup Trap Error Suppression** | Trap previously used uncaptured exit code
   > s or suppressed cleanup failures (‘\|\| true‘), masking sandbox teardown fai
```

```
> lures. | ‘cleanup_boxes()‘ captures ‘$?‘, checks if ‘isolate --cleanup‘ fail
> s, logs the failure, and sets ‘FAILED=1‘, guaranteeing a non-zero exit statu
> s (‘exit 1‘). |
30 | | **Filesystem Boundary Matching** | Only ran a negative write test and matched
> broad strings (‘\|error‘). Any syntax error or missing tool caused a false-p
> ositive pass. | **Dual Control Added**: <br>1. **Positive Control**: Writes
> to ‘/box/positive_probe.txt‘ and asserts readback. <br>2. **Negative Control
> **: Probes ‘/test_boundary_probe‘, asserts non-zero exit, verifies ‘EROFS‘ /
>  ‘Read-only file system‘, and queries container to ensure the file does not
> exist on disk. |
31 | | **Network Denial Indeterminacy** | Python interpreter crashes or syntax errors
>  could be misread as network isolation if the script failed before the socke
> t call. | Python probe now emits ‘PROBE_INITIATED‘ first. Catches ‘OSError‘,
>  logs exact ‘e.errno‘, and asserts ‘NETWORK_ISOLATED‘. If ‘PROBE_INITIATED‘
> is missing, the test fails immediately. |
32 | | **Memory Limit Verification** | Only checked ‘! echo "$MEM_OUT" \| grep -q "VU
> LNERABLE_MEM_OVERRUN"‘. A syntax error or immediate crash passed as a memory
>  limit enforcement. | Configured Isolate with ‘-M /tmp/isolate_mem_meta‘. Th
> e test now inspects the meta file for kernel cgroup OOM evidence: ‘status:SG
> ‘, ‘status:RE‘, ‘exitsig:9‘, or ‘cg-oom-killed:1‘. |
33 | | **CPU Timeout Unconditional Pass** | Contained an ‘else pass "Isolate CPU time
> out prevented runaway execution"‘ branch. If the runaway process exited with
>  0, it still reported PASS! | Configured Isolate with ‘-M /tmp/isolate_cpu_m
> eta‘. The test fails if exit is 0 (‘Runaway CPU process unexpectedly exited
> cleanly‘), and strictly matches ‘status:TO‘ or ‘exitsig:24/9‘. |
34 | | **Sandbox Directory Cleanup Distinction** | Docker CLI failure or SSH connecti
> on loss could be misinterpreted as directory absence (‘CLEAN‘). | Captured D
> ocker exit code ‘$DIR_STATUS_EXIT‘. Distinguishes ‘test -d‘ missing (‘CLEAN‘
> , exit 0), directory lingering (‘EXISTS‘), and Docker execution error (‘exit
>  > 0‘). |
35 | | **Residual Process Leakage** | Did not verify whether sandbox worker processes
>  lingered in memory after cleanup. | Added ‘pgrep -f "box-$BOX_ID"‘. Any lin
> gering processes immediately trigger ‘fail‘ and ‘FAILED=1‘. |
36 |
37 | ---
38 |
39 | ### 1.2 ‘deploy/judge0/verify-resource-accounting.sh‘
40 |
41 | | Prior Vulnerability / Flaw | Mechanism of False-Positive | Corrected Fail-Clos
> ed Implementation |
42 | | :--- | :--- | :--- |
43 | | **Single-Shot Submission Query** | Executed a single curl after a fixed sleep.
>  If the worker was slow, the submission was still in queue (Status 1/2), mis
> reporting the outcome. | Implemented an active polling loop querying ‘$ENDPO
> INT/submissions/$SUB_TOKEN‘ until terminal state (‘status.id >= 3‘) with a 3
> 0s timeout and 0.5s intervals. Fails closed on timeout. |
44 | | **Heuristic Process Tracking** | Sampled PIDs using ‘docker compose exec judge
> 0-workers pgrep -f "python3"‘, which could sample unrelated processes or mis
> s fast-spawning processes. | Directly inspects ‘/sys/fs/cgroup/memory/isolat
> e/box-*/cgroup.procs‘ on the host to obtain the exact kernel PID assigned to
>  the sandbox, then reads ‘/proc/$pid/cgroup‘ to prove hierarchy affinity. |
45 | | **Silent Telemetry Failures** | If no Isolate cgroup was found or peak usage r
> ead 0, the script continued and could print a PASS verdict. | Fails closed:
> If ‘PEAK_ISOLATE_USAGE -eq 0‘ or ‘CHILD_CGROUP_AFFINITY‘ is empty, ‘FAILED=1
> ‘ is set, printing ‘[FAIL]‘ and exiting 1. |
46 | | **Host OOM Events as Warnings Only** | Detected host kernel OOM killer events
> in ‘dmesg‘ were logged as ‘[WARNING]‘ without failing the probe. | Host OOM
> events (‘HOST_OOM_COUNT > 0‘) now invoke ‘fail‘ and set ‘FAILED=1‘, blocking
>  runner validation. |
47 | | **False Guarantee of 2100 MB Headroom** | Stated that calculated ~2100 MB head
> room "easily absorbs" sandbox loads, implying absolute host immunity from OO
> M. | Removed false guarantee. Clearly distinguishes configured container lim
```

```
     > its (4992 MB), actual measured consumption, and real-time host ‘MemAvailable
     > ‘. Added risk notice explaining host OOM dynamics under sibling cgroups. |
48 |
49 | ---
50 |
51 | ### 1.3 ‘deploy/judge0/verify-containment.sh‘
52 |
53 | | Prior Vulnerability / Flaw | Mechanism of False-Positive | Corrected Fail-Clos
     > ed Implementation |
54 | | :--- | :--- | :--- |
55 | | **Absence of Baseline Positive Control** | If the runner API was down, unauthe
     > nticated, or corrupted, all 25 tests could fail in ways that matched error c
     > onditions, masking complete runner failure. | **Added Section 0 Baseline Con
     > trol**: Submits valid code ‘print("BASELINE_VALID_EXECUTION")‘. Requires HTT
     > P 200/201, ‘status.id == 3‘, and exact stdout match. Aborts the entire suite
     >  immediately (‘exit 1‘) if baseline execution fails. |
56 | | **Curl Failure Masquerading as Containment** | ‘curl‘ failure fallback produce
     > d ‘{"status":{"id":13,"description":"Curl Error"}}‘. Tests matching Status 1
     > 3 or pattern ‘error‘ marked network failure as successful defense! | Curl fa
     > llback to JSON removed. If curl fails or returns an empty body, the test fai
     > ls immediately. Status 13 (‘Internal Error‘) is explicitly rejected across a
     > ll tests: infrastructure crashes are never treated as containment. |
57 | | **Permissive Status 13 Acceptance** | Tests 5, 9, 18, 19, 22, 24 accepted ‘exp
     > ected_status="3\|11\|12\|13"‘. | Removed ‘13‘ from all expected status strin
     > gs. Only legitimate sandbox outcomes (‘3‘ for trapped, ‘5‘ for TLE, ‘11/12‘
     > for OOM/runtime error) are accepted. |
58 | | **Overly Broad Pattern Matching** | Test 5 matched ‘"NETWORK_BLOCKED\|error\|i
     > nvalid"‘, and Test 19 matched ‘"\|error"‘. Any fatal syntax or runner error
     > matched ‘error‘ and passed. | Replaced with specific assertions: ‘"NETWORK_B
     > LOCKED\|Operation not permitted\|Network is unreachable"‘ and ‘"CPP_FORK_BOM
     > B_CONTAINED\|Resource temporarily unavailable\|BlockingIOError"‘. |
59 |
60 | ---
61 |
62 | ## 2. Table of Security Assertions & Failure Behaviors
63 |
64 | | Test & Vector | Assertion Under Test | Expected Evidence | Strict Failure Trig
     > ger |
65 | | :--- | :--- | :--- | :--- |
66 | | **Baseline Sanity** | Runner is alive, reachable, and correctly executing vali
     > d code. | ‘status.id: 3‘, stdout: ‘BASELINE_VALID_EXECUTION‘. | Empty curl,
     > non-zero curl, ‘status.id != 3‘, or mismatched stdout. Aborts suite. |
67 | | **Auth Guards (1-3)** | Unauthenticated or unauthorized requests to ‘/submissi
     > ons‘ and ‘/system_info‘ are blocked. | HTTP ‘401 Unauthorized‘ or ‘403 Forbi
     > dden‘. | Any HTTP 2xx, 3xx, or 5xx response. |
68 | | **Network Egress (4, 20, 25)** | Sandboxed processes (Python, C++, JS) cannot
     > connect to external IPs (e.g. ‘8.8.8.8:53‘). | Socket errno 101/113 (‘NETWOR
     > K_ISOLATED‘), ‘CPP_NETWORK_BLOCKED‘, ‘JS_NETWORK_BLOCKED‘. | Connection succ
     > ess (‘VULNERABLE_NETWORK_OPEN‘), silent timeout without probe initiation, or
     >  Status 13. |
69 | | **Client Override (5)** | Submissions with ‘enable_network: true‘ are blocked
     > when runner policy disables network overrides. | Rejection or sandbox denial
     >  without external connectivity. | Outbound connection opened or Status 13 ma
     > squerading as containment. |
70 | | **Cloud Metadata (6)** | Submissions cannot reach AWS/Azure/GCP metadata servi
     > ce (‘169.254.169.254‘). | ‘METADATA_BLOCKED‘ via isolated network namespace.
     >  | Metadata retrieved (‘VULNERABLE_METADATA_ACCESSED‘) or Status 13. |
71 | | **Fork Bomb (7, 19)** | Process creation is strictly capped (‘pids.max‘ or Iso
     > late process limits). | ‘FORK_BOMB_CONTAINED‘, ‘BlockingIOError‘, or ‘Resour
     > ce temporarily unavailable‘. | Fork bomb continues unhindered, host PID exha
     > ustion, or Status 13. |
72 | | **Heap Memory (8, 9, 18, 22, 24)** | Single-process and multi-process aggregat
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```
> e memory exhaustion is contained by cgroup memory limits. | Isolate cgroup O
> OM: ‘status:SG‘, ‘status:RE‘, ‘exitsig:9‘, or ‘Memory limit exceeded‘. | Zer
> o exit code, completion of full allocation (‘VULNERABLE_MEMORY_LEAK‘), host-
> level OOM, or Status 13. |
 73 | | **CPU Timeouts (10, 17, 21, 23)** | Infinite CPU loops are terminated at confi
> gured timeout. | Isolate meta ‘status:TO‘, ‘exitsig:24/9‘, or ‘Time Limit Ex
> ceeded‘. | Process exits with code 0, process continues running past wall-ti
> me limit, or Status 13. |
 74 | | **Filesystem Jail (11, 12)** | Root filesystem is read-only; disk writes are r
> estricted to ‘/box‘ quota. | ‘EROFS‘, ‘PASSWD_READONLY_CONTAINED‘, ‘DISK_LIM
> IT_CONTAINED‘ (‘SIGXFSZ‘). | File created on rootfs (‘/test_boundary_probe‘
> or ‘/etc/passwd‘), or quota bypassed. |
 75 | | **Secret Sniffing (13)** | Host environment variables, API tokens, and databas
> e credentials are not leaked into sandbox. | ‘ENV_SECRETS_SANITIZED‘ (no env
> ironment variables containing ‘AUTH‘, ‘PASS‘, ‘SECRET‘, ‘TOKEN‘). | Sandbox
> environment discovers secrets (‘VULNERABLE_SECRETS_FOUND‘). |
 76 | | **PID Namespace (14)** | Sandboxed process cannot view host or container proce
> sses in ‘/proc‘. | ‘PID_NAMESPACE_ISOLATED‘ (3 or fewer PIDs visible in ‘/pr
> oc‘). | Container or host PIDs visible (‘VULNERABLE_HOST_PIDS_VISIBLE > 3‘).
>  |
 77 | | **Privilege Escalation (15)** | Calling ‘setuid(0)‘ inside the sandbox fails.
> | ‘PRIVILEGE_ESCALATION_BLOCKED‘ (‘EPERM‘). | ‘setuid(0)‘ returns 0 (‘VULNER
> ABLE_SETUID_ROOT_SUCCESS‘). |
 78 | | **Raw Sockets (16)** | Creating raw network sockets (‘SOCK_RAW‘) fails due to
> dropped ‘CAP_NET_RAW‘. | ‘RAW_SOCKET_BLOCKED‘ (‘PermissionError‘). | Raw soc
> ket created (‘VULNERABLE_RAW_SOCKET_OPENED‘). |
 79 |
 80 | ---
 81 |
 82 | ## 3. Critical Analysis of Privileged Worker Risks
 83 |
 84 | A core architectural aspect of the Judge0 architecture is the ‘privileged: true‘
>  flag configured on the ‘judge0-workers‘ container in [‘deploy/judge0/docker
> -compose.yml‘](file:///d:/TestForge/deploy/judge0/docker-compose.yml#L55).
 85 |
 86 | ### 3.1 Exact Host Privileges Held by ‘judge0-workers‘
 87 | Running Docker with ‘privileged: true‘ disables almost all Linux container isola
> tion boundaries:
 88 | 1. **Raw Block Device Access:** All host block devices (‘/dev/sda‘, ‘/dev/sda1‘,
>  ‘/dev/nvme0n1‘, ‘/dev/vda‘) are exposed with read/write access directly ins
> ide the container filesystem under ‘/dev‘.
 89 | 2. **All Linux Capabilities Retained:** Full capability set (‘CAP_SYS_ADMIN‘, ‘C
> AP_SYS_RAWIO‘, ‘CAP_SYS_PTRACE‘, ‘CAP_DAC_OVERRIDE‘, ‘CAP_NET_ADMIN‘, ‘CAP_S
> YS_MODULE‘, etc.) is granted to root processes in the container.
 90 | 3. **Disabled Security Profiles:** Docker’s default AppArmor profile is disabled
>  (‘unconfined‘), and the default seccomp filter is disabled (‘unconfined‘).
 91 | 4. **Direct Host Cgroup Hierarchy Access:** Direct read-write access to host ‘/s
> ys/fs/cgroup‘.
 92 |
 93 | ### 3.2 Debunking Common Mitigations as Host Escape Protections
 94 | The project implements several layers of defense:
 95 | - Untrusted student code is executed by Isolate as an unprivileged user (UID ‘60
> 000+‘).
 96 | - The Isolate binary is owned by ‘root:root‘ with SUID permissions (‘mode 4755‘)
> .
 97 | - The Docker socket (‘/var/run/docker.sock‘) is **NOT** mounted into the worker
> container.
 98 |
 99 | **Why these mitigations do NOT neutralize container escape risk:**
100 | - **Isolate as the Single Point of Failure:** Isolate executes with root privile
> ges to configure namespaces, chroots, and cgroup controllers before dropping
>  privileges to UID ‘60000+‘. If an attacker discovers a vulnerability in Iso
```

```
> late (such as a namespace breakout, mount point traversal, file descriptor l
> eak, ptrace interaction bug, or SUID privilege escalation) OR a Linux kernel
>  zero-day (such as an exploit targeting user namespaces, io_uring, eBPF, or
> cgroup v1 controllers), the execution context transitions directly to contai
> ner root.
101 | - **Escape from Container = Full Host Compromise:** In an unprivileged container
> , escaping the sandbox still leaves the attacker constrained by container na
> mespaces, dropped capabilities, and AppArmor. However, in ‘judge0-workers‘ (
> ‘privileged: true‘), escaping the sandbox lands the attacker in a privileged
>  container context. From there, host compromise is trivial: the attacker can
>  directly mount ‘/dev/sda1‘ to a directory and write an SSH authorized key o
> r root cron job, write directly to raw disk blocks, or use ‘release_agent‘ i
> n ‘/sys/fs/cgroup‘ to execute arbitrary code as host root.
102 | - **Absence of ‘docker.sock‘ is Irrelevant:** While not mounting ‘docker.sock‘ p
> revents querying the Docker engine daemon, a privileged container has direct
>  kernel and hardware access that bypasses the Docker daemon entirely.
103 |
104 | ### 3.3 Host Denial-of-Service (DoS) Vectors
105 | 1. **Host RAM Exhaustion via Sibling Cgroups:**
106 |    - As verified in Section 9 of ‘verify-resource-accounting.sh‘, Isolate create
> s box cgroups directly under the host root hierarchy (‘/sys/fs/cgroup/memory
> /isolate/box-*‘), **NOT** nested inside the Docker worker container cgroup s
> lice.
107 |    - Therefore, Docker’s ‘mem_limit: 3072m‘ on ‘judge0-workers‘ does **NOT** bou
> nd Isolate sandboxes.
108 |    - Sandbox memory is bounded solely by Isolate’s internal ‘--cg-mem‘ limit (de
> fault 250 MB).
109 |    - If concurrency is increased beyond ‘COUNT=1‘ or if memory is consumed by ke
> rnel page cache, slab, and container processes, the host physical RAM (8 GB)
>  can be exhausted, triggering the Linux kernel host OOM killer.
110 | 2. **Host Disk & Inode Exhaustion:**
111 |    - While Isolate enforces a file size limit (‘fsize‘) inside ‘/box‘, files gen
> erated by compilers, runtime environments, and worker queues under ‘/tmp‘ or
>  ‘/var/local/lib/isolate‘ could exhaust available space or inodes on the 30
> GB root disk if automated cleanup fails.
112 |
113 | ---
114 |
115 | ## 4. Automated Verification Results
116 |
117 | The automated regression and audit test suite was executed locally:
118 | ‘‘‘powershell
119 | npm --prefix server run test -- tests/phase51C6FalsePositiveAudit.test.js tests/
> phase51C5FinalVerification.test.js tests/phase51C4BuildHardening.test.js tes
> ts/phase51C3Remediation.test.js tests/backendToRunnerIntegration.test.js tes
> ts/healthReadiness.test.js
120 | ‘‘‘
121 |
122 | ### Result Summary
123 | ‘‘‘
124 | PASS tests/phase51C6FalsePositiveAudit.test.js
125 | PASS tests/healthReadiness.test.js
126 | PASS tests/backendToRunnerIntegration.test.js
127 | PASS tests/phase51C4BuildHardening.test.js
128 | PASS tests/phase51C3Remediation.test.js
129 | PASS tests/phase51C5FinalVerification.test.js
130 |
131 | Test Suites: 6 passed, 6 total
132 | Tests:       112 passed, 112 total
133 | Snapshots:   0 total
134 | Time:        6.473 s
135 | ‘‘‘
```

```
136 |
137 | ---
138 |
139 | ## 5. Unresolved Risks & Clear Deployment Decision
140 |
141 | ### 5.1 Remaining Unresolved Operational Risks
142 | 1. **Live Cloud Environment Validation Required:**
143 |    All verifications performed to date are repository-level static and algorithm
      > ic validations. While all test scripts and Docker configurations have been a
      > udited and verified fail-closed, real kernel execution on the target Azure A
      > RM64 VM (‘Ubuntu 22.04.5 LTS‘, Kernel ‘6.8.0-1070-azure‘) is required to con
      > firm that the Azure-specific kernel behaves identically with cgroups v1 lega
      > cy controllers.
144 | 2. **Privileged Worker Attack Surface:**
145 |    Because ‘judge0-workers‘ requires ‘privileged: true‘ to operate Isolate’s cgr
      > oups and namespace setup, the runner VM must remain strictly **isolated** on
      >  the Azure virtual network, with no inbound access permitted from the public
      >  internet (accessible only from the TestForge backend via authenticated Cadd
      > y reverse proxy).
146 | 3. **Concurrency Restriction in Staging:**
147 |    Concurrency must strictly remain hardcoded to ‘COUNT=1‘ (‘MAX_QUEUE_SIZE=100‘
      > ) on the 8 GB / 2 vCPU VM to prevent sibling cgroup memory contention.
148 |
149 | ### 5.2 Build & Deployment Recommendation
150 | - **Verdict: PROCEED TO STAGING BUILD AND PROVISIONING TEST.**
151 | - **Justification:**
152 |   - All false-positive paths in ‘verify-cgroups-isolate.sh‘, ‘verify-resource-ac
      > counting.sh‘, and ‘verify-containment.sh‘ have been eradicated.
153 |   - Image architectures are locked to native ARM64 with fail-closed compose gate
      > s.
154 |   - Sudoers permissions and cron daemon wrappers are secured with restrictive ow
      > ner-only permissions.
155 |   - The TestForge backend connector implements circuit breaker, mutual authentic
      > ation, and strict status code handling.
156 | - The team may now proceed to build the ARM64 container images (‘testforge/compi
      > lers:v1.4.0-arm64‘ and ‘testforge/judge0:v1.13.1-arm64‘) and execute the ver
      > ification probes on the dedicated staging Azure VM.
157 |
```

2. Cgroups v1 & Isolate Operational Check Script

File Path: deploy/judge0/verify-cgroups-isolate.sh [SHELL SCRIPT (BASH)]

Hardened operational probe: non-zero cleanup trap, dual filesystem controls (positive & EROFS), socket errno validation, and Isolate meta checks.

```
#!/usr/bin/env bash
# ============================================================================
# TestForge Dedicated Execution Runner - Cgroups v1 & Isolate Operational Check
# Verifies host cgroup v1 hierarchy, controller writeability, and isolate sandbo
> x lifecycle
# Run on the Linux Runner VM before activating the runner
# ============================================================================

set -euo pipefail

echo "============================================================"
echo " TestForge Cgroups v1 & Isolate Operational Verification"
echo " Date: $(date -u)"
echo "============================================================"

FAILED=0

fail() {
  echo "  [FAIL] $1"
  FAILED=1
}

pass() {
  echo "  [PASS] $1"
}

# 1. Kernel Boot Parameter Check
echo ""
echo "â  â   1. Kernel Boot Parameters â  â  "
if grep -q "systemd.unified_cgroup_hierarchy=0" /proc/cmdline; then
  pass "systemd.unified_cgroup_hierarchy=0 present in /proc/cmdline"
else
  fail "Kernel did not boot with systemd.unified_cgroup_hierarchy=0 (cgroups v1
> legacy required)."
fi

# 2. Check cgroups v1 Controller Mounts
echo ""
echo "â  â   2. Controller Mounts & Hierarchy â  â  "
CONTROLLERS=("memory" "cpuacct" "cpuset")

for ctl in "${CONTROLLERS[@]}"; do
  # Check if mount point exists
  CTL_PATH=""
  if [ -d "/sys/fs/cgroup/$ctl" ]; then
    CTL_PATH="/sys/fs/cgroup/$ctl"
  elif [ -d "/sys/fs/cgroup/cpu,$ctl" ]; then
    CTL_PATH="/sys/fs/cgroup/cpu,$ctl"
```

```
  elif [ -d "/sys/fs/cgroup/$ctl,cpu" ]; then
    CTL_PATH="/sys/fs/cgroup/$ctl,cpu"
  fi

  if [ -n "$CTL_PATH" ]; then
    pass "Controller '$ctl' found at $CTL_PATH"

    # Check if writable
    TEST_DIR="$CTL_PATH/testforge_verify_probe_$$"
    if mkdir "$TEST_DIR" 2>/dev/null; then
      rmdir "$TEST_DIR"
      pass "Controller '$ctl' is writable (mkdir/rmdir successful)"
    else
      fail "Controller '$ctl' at $CTL_PATH is READ-ONLY or inaccessible"
    fi
  else
    fail "Required cgroup v1 controller '$ctl' is NOT mounted in /sys/fs/cgroup"
  fi
done

BOX_ID=98

cleanup_boxes() {
  local exit_code=$?
  if [ "$exit_code" -ne 0 ] && command -v docker &>/dev/null; then
    echo "  [CLEANUP TRAP] Cleaning up box $BOX_ID after non-zero exit ($exit_co
> de)..."
    if ! docker compose exec -T judge0-workers isolate --cleanup --box-id="$BOX_
> ID" --cg 2>&1; then
      echo "  [CLEANUP TRAP ERROR] isolate --cleanup failed during error recover
> y."
      FAILED=1
    fi
  fi
}
trap cleanup_boxes EXIT INT TERM ERR

echo ""
echo "â  â   3. DIRECT ISOLATE SANDBOX ENGINE TESTS (Inside judge0-workers Container
> ) â  â  "
if ! command -v docker &>/dev/null; then
  fail "Docker CLI is not installed or not in PATH."
else
  # Check if container is running
  if docker compose ps --services --filter "status=running" 2>/dev/null | grep -
> q "judge0-workers"; then
    pass "judge0-workers container is running"

    # 3.1 Test isolate binary existence, SUID permissions, and helper files
    if docker compose exec -T judge0-workers test -u /usr/local/bin/isolate && \
       docker compose exec -T judge0-workers test -x /usr/local/bin/isolate; the
> n
      pass "isolate binary installed with SUID root (mode 4755)"
    else
      fail "isolate binary is missing or lacks SUID root permission"
    fi

    if docker compose exec -T judge0-workers test -f /usr/local/etc/isolate; the
> n
      pass "isolate configuration /usr/local/etc/isolate is present"
    else
      fail "isolate configuration /usr/local/etc/isolate is missing"
```

```
102 |     fi
103 |
104 |     ENV_CHECK=$(docker compose exec -T judge0-workers /usr/local/bin/isolate-che
    > ck-environment 2>&1 || echo "ERROR")
105 |     if [ "$ENV_CHECK" != "ERROR" ]; then
106 |       pass "isolate-check-environment executed successfully"
107 |     else
108 |       fail "isolate-check-environment failed: $ENV_CHECK"
109 |     fi
110 |
111 |     # 3.2 Ensure Box 98 is in clean state before initialization
112 |     if docker compose exec -T judge0-workers test -d "/var/local/lib/isolate/$BO
    > X_ID"; then
113 |       docker compose exec -T judge0-workers isolate --cleanup --box-id=$BOX_ID -
    > -cg
114 |     fi
115 |
116 |     INIT_OUT=$(docker compose exec -T judge0-workers isolate --init --box-id=$BO
    > X_ID --cg 2>&1)
117 |     INIT_EXIT=$?
118 |     if [ "$INIT_EXIT" -eq 0 ] && echo "$INIT_OUT" | grep -q "/var/local/lib/isol
    > ate"; then
119 |       pass "isolate --init --cg succeeded (Box $BOX_ID initialized at $INIT_OUT)
    > "
120 |     else
121 |       fail "isolate --init --cg failed (exit $INIT_EXIT): $INIT_OUT"
122 |     fi
123 |
124 |     # Verify physical sandbox directory and cgroup existence
125 |     if docker compose exec -T judge0-workers test -d "/var/local/lib/isolate/$BO
    > X_ID/box"; then
126 |       pass "Sandbox directory /var/local/lib/isolate/$BOX_ID/box verified presen
    > t"
127 |     else
128 |       fail "Sandbox directory /var/local/lib/isolate/$BOX_ID/box was not created
    > "
129 |     fi
130 |
131 |     if docker compose exec -T judge0-workers test -d "/sys/fs/cgroup/memory/isol
    > ate/box-$BOX_ID" || \
132 |        docker compose exec -T judge0-workers test -d "/sys/fs/cgroup/memory/box-
    > $BOX_ID"; then
133 |       pass "Isolate memory cgroup box directory verified present"
134 |     else
135 |       fail "Isolate memory cgroup box directory was not created"
136 |     fi
137 |
138 |     # 3.3 Test Execution Sanity (Positive Control: Echo inside Isolate)
139 |     RUN_OUT=$(docker compose exec -T judge0-workers isolate --box-id=$BOX_ID --c
    > g --cg-mem=64000 --run -- /bin/echo "TESTFORGE_ISOLATE_OK" 2>&1)
140 |     RUN_EXIT=$?
141 |     if [ "$RUN_EXIT" -eq 0 ] && echo "$RUN_OUT" | grep -q "TESTFORGE_ISOLATE_OK"
    > ; then
142 |       pass "Positive Control: isolate basic sandboxed execution succeeded"
143 |     else
144 |       fail "Positive Control: isolate basic execution failed (exit $RUN_EXIT): $
    > RUN_OUT"
145 |     fi
146 |
147 |     # 3.4 Filesystem Boundary Tests: Positive (/box write) & Negative (read-only
    >  rootfs)
148 |     # Positive Control: Writing inside /box must succeed
```

```
    FS_POS_OUT=$(docker compose exec -T judge0-workers isolate --box-id=$BOX_ID
> --cg --run -- /bin/bash -c 'echo "OK" > /box/positive_probe.txt && cat /box/
> positive_probe.txt' 2>&1)
    FS_POS_EXIT=$?
    if [ "$FS_POS_EXIT" -eq 0 ] && echo "$FS_POS_OUT" | grep -q "OK"; then
      pass "Positive control: Sandbox allowed write within /box"
    else
      fail "Filesystem Positive Control: Writing inside sandbox jail failed (exi
> t $FS_POS_EXIT): $FS_POS_OUT"
    fi

    # Negative Control: Writing to rootfs outside /box must fail specifically wi
> th EROFS / Read-only file system
    FS_NEG_OUT=$(docker compose exec -T judge0-workers isolate --box-id=$BOX_ID
> --cg --run -- /bin/touch /test_boundary_probe 2>&1)
    FS_NEG_EXIT=$?
    if [ "$FS_NEG_EXIT" -ne 0 ] && echo "$FS_NEG_OUT" | grep -qiE "Read-only fil
> e system|EROFS"; then
      # Verify target file does not exist in container
      if docker compose exec -T judge0-workers test ! -e "/test_boundary_probe";
>  then
        pass "Isolate filesystem boundary enforced (Read-only rootfs strictly en
> forced; EROFS on / write, exit $FS_NEG_EXIT)"
      else
        fail "Filesystem Breach: File /test_boundary_probe was written to rootfs
> !"
      fi
    else
      fail "Filesystem Boundary Failed: Expected EROFS non-zero failure, got exi
> t $FS_NEG_EXIT, output: $FS_NEG_OUT"
    fi

    # 3.5 Network Denial Enforcement (Zero Egress inside Isolate)
    NET_OUT=$(docker compose exec -T judge0-workers isolate --box-id=$BOX_ID --c
> g --run -- /usr/bin/python3 -c '
import socket
import sys
sys.stdout.write("PROBE_INITIATED\n")
sys.stdout.flush()
s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.settimeout(2.0)
try:
    s.connect(("8.8.8.8", 53))
    sys.stdout.write("VULNERABLE_NETWORK_OPEN\n")
except OSError as e:
    sys.stdout.write("NETWORK_ISOLATED:" + str(e.errno) + ":" + type(e).__name__
>  + "\n")
except Exception as e:
    sys.stdout.write("UNEXPECTED_EXCEPTION:" + type(e).__name__ + "\n")
' 2>&1)
    NET_EXIT=$?
    if ! echo "$NET_OUT" | grep -q "PROBE_INITIATED"; then
      fail "Network probe did not initiate cleanly: Python interpreter failed ($
> NET_OUT)"
    elif echo "$NET_OUT" | grep -q "VULNERABLE_NETWORK_OPEN"; then
      fail "Network Isolation Breach: Outbound socket connection succeeded!"
    elif echo "$NET_OUT" | grep -q "NETWORK_ISOLATED"; then
      pass "Isolate network isolation enforced (Outbound socket connection block
> ed by network namespace)"
    else
      fail "Network Test Indeterminate: Unexpected output: $NET_OUT"
    fi
```

```
197 |
198 |     # 3.6 Memory Limit Enforcement (Cgroup OOM containment inside Isolate)
199 |     docker compose exec -T judge0-workers rm -f /tmp/isolate_mem_meta
200 |     MEM_OUT=$(docker compose exec -T judge0-workers isolate --box-id=$BOX_ID --c
    > g --cg-mem=32000 -M /tmp/isolate_mem_meta --run -- /usr/bin/python3 -c '
201 | import sys
202 | sys.stdout.write("ALLOC_START\n")
203 | sys.stdout.flush()
204 | chunks = [bytearray(10 * 1024 * 1024) for _ in range(8)]
205 | sys.stdout.write("ALLOC_FINISHED\n")
206 | ' 2>&1)
207 |     MEM_EXIT=$?
208 |     MEM_META=$(docker compose exec -T judge0-workers cat /tmp/isolate_mem_meta 2
    > >/dev/null || echo "")
209 |
210 |     if echo "$MEM_OUT" | grep -q "ALLOC_FINISHED"; then
211 |       fail "Memory Limit Breach: Process allocated past 32MB limit without termi
    > nation!"
212 |     elif [ "$MEM_EXIT" -ne 0 ] && echo "$MEM_META" | grep -qiE "status:SG|status
    > :RE|exitsig:9|cg-oom-killed:1|Memory limit exceeded"; then
213 |       pass "Isolate cgroup memory limit enforced (Kernel cgroup OOM killed proce
    > ss; meta: $(echo "$MEM_META" | tr '\n' ' '))"
214 |     else
215 |       fail "Memory Limit Indeterminate: Exit code $MEM_EXIT, Meta: $MEM_META, Ou
    > tput: $MEM_OUT"
216 |     fi
217 |
218 |     # 3.7 CPU Timeout Enforcement inside Isolate
219 |     docker compose exec -T judge0-workers rm -f /tmp/isolate_cpu_meta
220 |     CPU_OUT=$(docker compose exec -T judge0-workers isolate --box-id=$BOX_ID --c
    > g --time=1 --wall-time=2 -M /tmp/isolate_cpu_meta --run -- /bin/bash -c 'whi
    > le true; do :; done' 2>&1)
221 |     CPU_EXIT=$?
222 |     CPU_META=$(docker compose exec -T judge0-workers cat /tmp/isolate_cpu_meta 2
    > >/dev/null || echo "")
223 |
224 |     if [ "$CPU_EXIT" -eq 0 ]; then
225 |       fail "Runaway CPU process unexpectedly exited cleanly with exit code 0!"
226 |     elif echo "$CPU_META" | grep -qiE "status:TO|exitsig:9|exitsig:24|Time limit
    >  exceeded"; then
227 |       pass "Isolate CPU time limit enforced (Runaway loop terminated by Isolate
    > timer; meta: $(echo "$CPU_META" | tr '\n' ' '))"
228 |     else
229 |       fail "CPU Timeout Indeterminate: Exit $CPU_EXIT, Meta: $CPU_META, Output:
    > $CPU_OUT"
230 |     fi
231 |
232 |     # 3.8 Test Box Cleanup & Cgroup Removal (Strict Non-Zero Validation)
233 |     CLEAN_OUT=$(docker compose exec -T judge0-workers isolate --cleanup --box-id
    > =$BOX_ID --cg 2>&1)
234 |     CLEAN_EXIT=$?
235 |     if [ "$CLEAN_EXIT" -eq 0 ]; then
236 |       pass "isolate --cleanup command exited successfully (code 0)"
237 |     else
238 |       fail "isolate --cleanup command failed with exit code $CLEAN_EXIT: $CLEAN_
    > OUT"
239 |     fi
240 |
241 |     # Verify physical cleanup of sandbox directory (Distinguish missing director
    > y from Docker failure)
242 |     DIR_CHECK=$(docker compose exec -T judge0-workers bash -c 'test -d "/var/loc
    > al/lib/isolate/'$BOX_ID'" && echo "EXISTS" || echo "CLEAN"')
```

```
    DIR_STATUS_EXIT=$?
    if [ "$DIR_STATUS_EXIT" -ne 0 ]; then
      fail "Failed to query sandbox directory state (Docker exit $DIR_STATUS_EXI
> T)"
    elif [ "$DIR_CHECK" = "CLEAN" ]; then
      pass "Sandbox directory /var/local/lib/isolate/$BOX_ID removed cleanly (bo
> x directory already cleaned up)"
    else
      fail "Sandbox directory /var/local/lib/isolate/$BOX_ID leaked (still exist
> s after cleanup)"
    fi

    # Verify cgroup cleanup (Distinguish missing cgroup from Docker failure)
    CG_CHECK=$(docker compose exec -T judge0-workers bash -c '([ -d "/sys/fs/cgr
> oup/memory/isolate/box-'$BOX_ID'" ] || [ -d "/sys/fs/cgroup/memory/box-'$BOX
> _ID'" ]) && echo "EXISTS" || echo "CLEAN"')
    CG_STATUS_EXIT=$?
    if [ "$CG_STATUS_EXIT" -ne 0 ]; then
      fail "Failed to query cgroup directory state (Docker exit $CG_STATUS_EXIT)
> "
    elif [ "$CG_CHECK" = "CLEAN" ]; then
      pass "Isolate memory cgroup box directory removed cleanly"
    else
      fail "Isolate memory cgroup box directory leaked (still exists after clean
> up)"
    fi

    # Verify no residual processes in system belonging to test box
    PROC_CHECK=$(docker compose exec -T judge0-workers bash -c 'pgrep -f "box-'$
> BOX_ID'" 2>/dev/null || echo "NONE"')
    if [ "$PROC_CHECK" = "NONE" ] || [ -z "$PROC_CHECK" ]; then
      pass "No residual processes found for Box $BOX_ID"
    else
      fail "Lingering processes detected for box-$BOX_ID: $PROC_CHECK"
    fi
  else
    fail "judge0-workers container is not running (run 'docker compose up -d' fi
> rst)."
  fi
fi

echo ""
echo "============================================================"
if [ "$FAILED" -eq 0 ]; then
  echo " VERDICT: ALL CGROUPS V1 & ISOLATE CHECKS PASSED [READY]"
  exit 0
else
  echo " VERDICT: CGROUPS V1 / ISOLATE VERIFICATION FAILED [FAIL-CLOSED]"
  echo " Correct kernel parameters and mount permissions before activating runne
> r."
  exit 1
fi
```

3. Host Resource & Cgroup Accounting Probe

File Path: deploy/judge0/verify-resource-accounting.sh [SHELL SCRIPT (BASH)]

Resource accounting probe: active polling loop until terminal state, kernel PID tracking via cgroup.procs, dmesg host OOM blocking, and honest headroom

```
#!/usr/bin/env bash
# ============================================================================
# TestForge Dedicated Execution Runner - Resource & Cgroup Accounting Probe
# Target: Native ARM64 Linux VM with cgroups v1
# Measures:
#   1. Worker container cgroup memory usage (Docker slice)
#   2. Host-level Isolate box cgroup memory usage (/sys/fs/cgroup/memory/isolate )
#   3. Total host memory & unallocated headroom (distinguishing limits vs RSS)
#   4. Determines whether Isolate child processes are nested or root siblings
#   5. Strictly fails closed if telemetry is missing, host OOM occurs, or orphans linger
# ============================================================================

set -euo pipefail

ENDPOINT="${1:-http://127.0.0.1:2358}"
TOKEN="${2:-$(grep -E '^AUTHN_TOKEN=' /opt/testforge-judge0/judge0.conf 2>/dev/null | cut -d'=' -f2 || grep -E '^AUTHN_TOKEN=' ./judge0.conf 2>/dev/null | cut -d'=' -f2 || echo '')}"

HEADERS=(-H "Content-Type: application/json")
if [ -n "$TOKEN" ]; then
  HEADERS+=(-H "X-Auth-Token: $TOKEN")
fi

FAILED=0

fail() {
  echo "  [FAIL] $1" >&2
  FAILED=1
}

echo "========================================================="
echo " TestForge Cgroup Memory Accounting & Headroom Probe   "
echo " Date:         $(date -u)"
echo " Endpoint:     $ENDPOINT"
echo "========================================================="

if ! command -v docker &>/dev/null; then
  echo "[FATAL] Docker CLI not found. Run this on the Linux runner VM." >&2
  exit 1
fi

WORKER_CID=$(docker compose ps -q judge0-workers 2>/dev/null || docker ps -q --filter "name=testforge-judge0-workers" 2>/dev/null || echo "")

if [ -z "$WORKER_CID" ]; then
```

```
  echo "[FATAL] judge0-workers container is not running. Start stack with 'docke
> r compose up -d' first." >&2
  exit 1
fi

echo "Worker Container ID: $WORKER_CID"

# 1. Inspect Host Physical Memory (Baseline)
echo ""
echo "â  â   1. Host Physical Memory Snapshot (Pre-Execution Baseline) â  â  "
if [ ! -r /proc/meminfo ]; then
  echo "[FATAL] /proc/meminfo is unreadable." >&2
  exit 1
fi

TOTAL_RAM_KB=$(grep MemTotal /proc/meminfo | awk '{print $2}')
AVAIL_RAM_KB_BEFORE=$(grep MemAvailable /proc/meminfo | awk '{print $2}')
TOTAL_RAM_MB=$((TOTAL_RAM_KB / 1024))
AVAIL_RAM_MB_BEFORE=$((AVAIL_RAM_KB_BEFORE / 1024))
echo "  Total Host RAM:              ${TOTAL_RAM_MB} MB"
echo "  Available Host RAM (Before): ${AVAIL_RAM_MB_BEFORE} MB"

# 2. Locate Docker Worker Container Cgroup Path
echo ""
echo "â  â   2. Docker Worker Cgroup Path â  â  "
DOCKER_CG_PATH=""
POSSIBLE_PATHS=(
  "/sys/fs/cgroup/memory/docker/$WORKER_CID"
  "/sys/fs/cgroup/memory/system.slice/docker-$WORKER_CID.scope"
  "/sys/fs/cgroup/memory/docker.slice/docker-$WORKER_CID.scope"
)

for p in "${POSSIBLE_PATHS[@]}"; do
  if [ -d "$p" ]; then
    DOCKER_CG_PATH="$p"
    break
  fi
done

DOCKER_BASELINE_USAGE=0
if [ -n "$DOCKER_CG_PATH" ] && [ -r "$DOCKER_CG_PATH/memory.usage_in_bytes" ]; t
> hen
  echo "  Worker Cgroup Found: $DOCKER_CG_PATH"
  DOCKER_BASELINE_USAGE=$(cat "$DOCKER_CG_PATH/memory.usage_in_bytes" 2>/dev/nul
> l || echo 0)
  DOCKER_CG_LIMIT=$(cat "$DOCKER_CG_PATH/memory.limit_in_bytes" 2>/dev/null || e
> cho 0)
  echo "  Baseline Worker Memory Usage: $((DOCKER_BASELINE_USAGE / 1024 / 1024))
>  MB"
  echo "  Configured Worker Limit:      $((DOCKER_CG_LIMIT / 1024 / 1024)) MB"
else
  fail "Worker container memory cgroup path not found or unreadable under /sys/f
> s/cgroup/memory"
fi

# 3. Locate Host Isolate Cgroup Path & Hierarchy Settings
echo ""
echo "â  â   3. Host Isolate Cgroup Path & Hierarchy Settings â  â  "
ISOLATE_CG_BASE="/sys/fs/cgroup/memory/isolate"
HOST_ROOT_CG="/sys/fs/cgroup/memory"

if [ -f "$HOST_ROOT_CG/memory.use_hierarchy" ]; then
```

```
  ROOT_HIERARCHY=$(cat "$HOST_ROOT_CG/memory.use_hierarchy")
  echo "  Host Root memory.use_hierarchy: $ROOT_HIERARCHY"
else
  fail "Host root memory.use_hierarchy unreadable"
fi

if [ -d "$ISOLATE_CG_BASE" ]; then
  echo "  Host Isolate Cgroup Base: $ISOLATE_CG_BASE"
  if [ -f "$ISOLATE_CG_BASE/memory.use_hierarchy" ]; then
    ISO_HIERARCHY=$(cat "$ISOLATE_CG_BASE/memory.use_hierarchy")
    echo "  Isolate Base memory.use_hierarchy: $ISO_HIERARCHY"
  fi
else
  echo "  [NOTICE] $ISOLATE_CG_BASE does not exist yet (will be initialized on f
> irst submission)."
fi

# 4. Dispatch Memory Test Payload (150 MB Allocation)
echo ""
echo "â  â   4. Dispatching Memory Accounting Test (150 MB Allocation) â  â  "
TEST_CODE='
import time
# Allocate 150MB of memory and hold for 3 seconds
chunks = [b"M" * (10 * 1024 * 1024) for _ in range(15)]
time.sleep(3)
print("ALLOCATION_HOLD_COMPLETE")
'

PAYLOAD=$(jq -n \
  --arg code "$TEST_CODE" \
  '{source_code: $code, language_id: 71, cpu_time_limit: 5.0, wall_time_limit: 8
> .0, memory_limit: 256000, enable_network: false}')

RESPONSE=$(curl -s -X POST "${HEADERS[@]}" -d "$PAYLOAD" "$ENDPOINT/submissions"
>  || echo "{}")
SUB_TOKEN=$(echo "$RESPONSE" | jq -r '.token // empty')

if [ -z "$SUB_TOKEN" ]; then
  echo "[FATAL] Failed to queue submission: $RESPONSE" >&2
  exit 1
fi
echo "  Submission Queued. Token: $SUB_TOKEN"

# 5. Active Telemetry Sampling During Execution
echo "  Actively sampling cgroup usage & tracking kernel PIDs during execution..
> ."
PEAK_DOCKER_USAGE=0
PEAK_ISOLATE_USAGE=0
CHILD_CGROUP_AFFINITY=""
DISCOVERED_BOX_PATH=""
DISCOVERED_PID=""
FAILCNT_VALUE=0

# Sample over execution window (up to ~6 seconds, sampling every 200ms)
for _ in $(seq 1 30); do
  sleep 0.2

  # Sample Docker cgroup usage
  if [ -n "$DOCKER_CG_PATH" ] && [ -f "$DOCKER_CG_PATH/memory.usage_in_bytes" ];
>  then
    CURR_DOCKER=$(cat "$DOCKER_CG_PATH/memory.usage_in_bytes" 2>/dev/null || ech
> o 0)
```

```
    if [ "$CURR_DOCKER" -gt "$PEAK_DOCKER_USAGE" ]; then
      PEAK_DOCKER_USAGE=$CURR_DOCKER
    fi
  fi

  # Sample Isolate cgroup usage and inspect cgroup.procs
  if [ -d "$ISOLATE_CG_BASE" ]; then
    for box_dir in "$ISOLATE_CG_BASE"/box-*; do
      if [ -d "$box_dir" ]; then
        DISCOVERED_BOX_PATH="$box_dir"

        # Read memory usage
        if [ -f "$box_dir/memory.usage_in_bytes" ]; then
          CURR_ISO=$(cat "$box_dir/memory.usage_in_bytes" 2>/dev/null || echo 0)
          if [ "$CURR_ISO" -gt "$PEAK_ISOLATE_USAGE" ]; then
            PEAK_ISOLATE_USAGE=$CURR_ISO
          fi
        fi

        # Track exact kernel PID from cgroup.procs
        if [ -f "$box_dir/cgroup.procs" ]; then
          BOX_PIDS=$(cat "$box_dir/cgroup.procs" 2>/dev/null || echo "")
          if [ -n "$BOX_PIDS" ]; then
            for pid in $BOX_PIDS; do
              if [ -r "/proc/$pid/cgroup" ]; then
                DISCOVERED_PID="$pid"
                CHILD_CGROUP_AFFINITY=$(grep "memory:" "/proc/$pid/cgroup" 2>/de
> v/null || echo "")
                break
              fi
            done
          fi
        fi

        # Check failcnt if available
        if [ -f "$box_dir/memory.failcnt" ]; then
          FAILCNT_VALUE=$(cat "$box_dir/memory.failcnt" 2>/dev/null || echo 0)
        fi
      fi
    done
  fi
done

# 6. Active Polling for Terminal State (Timeout: 30 seconds)
echo ""
echo "â  â   5. Polling Submission Status Until Terminal State â  â  "
TERMINAL_REACHED=false
POLL_TIMEOUT=30
ELAPSED=0
STATUS_ID=0
STATUS_DESC="Unknown"
RESULT_BODY="{}"

while [ "$ELAPSED" -lt "$POLL_TIMEOUT" ]; do
  RESULT_BODY=$(curl -s "$ENDPOINT/submissions/$SUB_TOKEN?base64_encoded=false"
> || echo "{}")
  STATUS_ID=$(echo "$RESULT_BODY" | jq -r '.status.id // 0')
  STATUS_DESC=$(echo "$RESULT_BODY" | jq -r '.status.description // "Unknown"')

  # Terminal states in Judge0: status.id >= 3 (3=Accepted, 4=Wrong Answer, 5=TLE
> , 6=Compilation Error, etc.)
  if [ "$STATUS_ID" -ge 3 ]; then
```

```
215 |     TERMINAL_REACHED=true
216 |     break
217 |   fi
218 |
219 |   sleep 0.5
220 |   ELAPSED=$((ELAPSED + 1))
221 | done
222 |
223 | if [ "$TERMINAL_REACHED" = false ]; then
224 |   fail "Submission $SUB_TOKEN timed out after ${POLL_TIMEOUT}s without reaching
    > terminal state (last status: $STATUS_ID - $STATUS_DESC)"
225 | else
226 |   echo "  Submission Resolved: Status $STATUS_ID ($STATUS_DESC)"
227 | fi
228 |
229 | # 7. Evaluate Telemetry Validity (Fail Closed if Telemetry Missing)
230 | echo ""
231 | echo "â  â   6. Telemetry & Accounting Verification â  â  "
232 | if [ "$PEAK_ISOLATE_USAGE" -eq 0 ]; then
233 |   fail "Missing Isolate cgroup telemetry: peak usage was 0 bytes across all /sys
    > /fs/cgroup/memory/isolate/box-* hierarchies"
234 | else
235 |   echo "  Peak Isolate Box Memory Usage: $((PEAK_ISOLATE_USAGE / 1024 / 1024)) M
    > B"
236 | fi
237 |
238 | if [ -z "$CHILD_CGROUP_AFFINITY" ]; then
239 |   fail "Failed to verify child process cgroup affinity: no kernel PID detected i
    > n isolate cgroup.procs"
240 | else
241 |   echo "  Discovered Kernel PID:         $DISCOVERED_PID"
242 |   echo "  Child Memory Cgroup Path:      $CHILD_CGROUP_AFFINITY"
243 | fi
244 |
245 | if [ -n "$DISCOVERED_BOX_PATH" ] && [ -f "$DISCOVERED_BOX_PATH/memory.oom_contro
    > l" ]; then
246 |   echo "  Box OOM Control Settings:"
247 |   sed 's/^/    /' "$DISCOVERED_BOX_PATH/memory.oom_control"
248 | fi
249 | echo "  Box memory.failcnt:            $FAILCNT_VALUE"
250 |
251 | # Check kernel dmesg for host-level OOM invocations (Release Blocker if > 0)
252 | echo ""
253 | echo "â  â   7. Host OOM Killer & Stability Audit â  â  "
254 | HOST_OOM_COUNT=$(dmesg 2>/dev/null | grep -iE 'oom-killer|out of memory|killed p
    > rocess' | wc -l || echo 0)
255 | if [ "$HOST_OOM_COUNT" -gt 0 ]; then
256 |   fail "Host kernel OOM events detected in dmesg ($HOST_OOM_COUNT events)! Poten
    > tial host-level OOM invocation."
257 | else
258 |   echo "  Host-level OOM killer invocations: 0 (No host kernel OOM events detect
    > ed)."
259 | fi
260 |
261 | # 8. Post-Execution Memory & Residual Process Audit
262 | echo ""
263 | echo "â  â   8. Post-Execution Cgroup Cleanup Verification â  â  "
264 | AVAIL_RAM_KB_AFTER=$(grep MemAvailable /proc/meminfo | awk '{print $2}')
265 | AVAIL_RAM_MB_AFTER=$((AVAIL_RAM_KB_AFTER / 1024))
266 | echo "  Available Host RAM (After):  ${AVAIL_RAM_MB_AFTER} MB"
267 |
268 | CLEANUP_OK=true
```

```
269 | if [ -d "$ISOLATE_CG_BASE" ]; then
270 |   ACTIVE_BOX_DIRS=$(find "$ISOLATE_CG_BASE" -mindepth 1 -maxdepth 1 -type d -nam
    > e "box-*" 2>/dev/null || echo "")
271 |   if [ -n "$ACTIVE_BOX_DIRS" ]; then
272 |     for bdir in $ACTIVE_BOX_DIRS; do
273 |       if [ -f "$bdir/cgroup.procs" ]; then
274 |         PROCS=$(cat "$bdir/cgroup.procs" 2>/dev/null || echo "")
275 |         if [ -n "$PROCS" ]; then
276 |           fail "Lingering active kernel processes found in $bdir cgroup.procs: $
    > PROCS"
277 |           CLEANUP_OK=false
278 |         fi
279 |       fi
280 |     done
281 |   fi
282 | fi
283 |
284 | # Verify no orphaned python3 or isolate execution processes linger
285 | ORPHAN_PROCS=$(pgrep -f "ALLOCATION_HOLD_COMPLETE" 2>/dev/null || echo "")
286 | if [ -n "$ORPHAN_PROCS" ]; then
287 |   fail "Orphaned test payload processes detected on host: $ORPHAN_PROCS"
288 | fi
289 |
290 | if [ "$CLEANUP_OK" = true ] && [ -z "$ORPHAN_PROCS" ]; then
291 |   echo "  Cgroup Cleanup: PASS (No orphan or zombie processes in isolate cgroups
    > )."
292 | fi
293 |
294 | # 9. Architectural Accounting Analysis & Honest Headroom Calculation
295 | echo ""
296 | echo "â  â   9. Cgroup Hierarchy & Realistic Memory Headroom Analysis â  â  "
297 | if echo "$CHILD_CGROUP_AFFINITY" | grep -q "docker"; then
298 |   echo "  [NESTED ACCOUNTING PROVEN]"
299 |   echo "  Isolate processes run inside the Docker container cgroup slice."
300 |   echo "  Docker 'mem_limit: 3072m' strictly bounds both worker and sandbox proc
    > esses."
301 | else
302 |   echo "  [ROOT SIBLING ACCOUNTING PROVEN]"
303 |   echo "  Isolate processes run directly under host root /sys/fs/cgroup/memory/i
    > solate/box-<id>."
304 |   echo "  CRITICAL ARCHITECTURAL REALITY:"
305 |   echo "  - Docker container 'mem_limit: 3072m' DOES NOT constrain Isolate child
    >  processes."
306 |   echo "  - Sandbox memory is bounded exclusively by Isolate's internal --cg-mem
    >  limit (256000 KB)."
307 |   echo "  - Sibling cgroups draw directly from host physical RAM alongside Docke
    > r containers."
308 |   echo "  - Concurrency is hardcoded to COUNT=1 in staging (max 250 MB sandbox l
    > oad)."
309 | fi
310 |
311 | echo ""
312 | echo "  Host Memory Budget Verification (8192 MB Physical RAM):"
313 | echo "  - Host OS & Kernel Baseline: ~1100 MB"
314 | echo "  - judge0-workers:           3072 MB"
315 | echo "  - judge0-server:            1024 MB"
316 | echo "  - judge0-db:                 512 MB"
317 | echo "  - judge0-redis:              384 MB"
318 | echo "  ------------------------------------------------------"
319 | echo "  - Total Container Allocation: 4992 MB (Reduced from 6528 MB by 1536 MB)"
320 | echo "  - Unallocated Host Headroom:  ~2100 MB buffer"
321 | echo "  - Isolate Per-Box Cap:         250 MB"
```

```
echo "  - Active Concurrency:            1 (COUNT=1 enforced)"
echo "  ------------------------------------------------------"
echo "  IMPORTANT RISK NOTICE (No absolute guarantee):"
echo "  - Mathematical difference (8192 - 4992 = 3200 MB, with ~2100 MB buffer)
> is a static calculation, NOT guaranteed safety."
echo "  - Actual host safety depends on real container RSS, Linux kernel slab/pa
> gecache, and active concurrency."
echo "  - Current Available Host RAM: ${AVAIL_RAM_MB_AFTER} MB"
echo "  - If worker concurrency is raised or sibling cgroups balloon, host OOM k
> iller can be triggered."
|
echo ""
echo "======================================================="
if [ "$FAILED" -eq 0 ]; then
  echo " PROBE RESULT: PASS (Resource accounting verified; telemetry valid)."
  exit 0
else
  echo " PROBE RESULT: FAIL (Missing telemetry, cgroup failure, or orphan proces
> ses)."
  exit 1
fi
|
```

4. Hostile-Code Containment Test Suite

File Path: deploy/judge0/verify-containment.sh [SHELL SCRIPT (BASH)]

Containment test suite: positive baseline control (sanity check), rejection of Status 13 (Internal Error), strict exit status matching, and cleanup.

```
#!/usr/bin/env bash
# ================================================================================
# TestForge Dedicated Execution Runner - Hostile-Code Containment Test Suite
# Target: Dedicated Linux VM with Judge0 CE v1.13.1
# Usage: ./verify-containment.sh [OPTIONAL_ENDPOINT] [OPTIONAL_AUTHN_TOKEN] [OPTIONAL_AUTHZ_TOKEN]
# ================================================================================

set -euo pipefail

ENDPOINT="${1:-http://127.0.0.1:2358}"
AUTHN_TOKEN="${2:-$(grep -E '^AUTHN_TOKEN=' /opt/testforge-judge0/judge0.conf 2>/dev/null | cut -d'=' -f2 || grep -E '^AUTHN_TOKEN=' ./judge0.conf 2>/dev/null | cut -d'=' -f2 || echo '')}"
AUTHZ_TOKEN="${3:-$(grep -E '^AUTHZ_TOKEN=' /opt/testforge-judge0/judge0.conf 2>/dev/null | cut -d'=' -f2 || grep -E '^AUTHZ_TOKEN=' ./judge0.conf 2>/dev/null | cut -d'=' -f2 || echo '')}"

HEADERS=(-H "Content-Type: application/json")
if [ -n "$AUTHN_TOKEN" ]; then
  HEADERS+=(-H "X-Auth-Token: $AUTHN_TOKEN")
fi

echo "============================================================"
echo " TestForge Hostile-Code Containment Verification Suite "
echo " Endpoint: $ENDPOINT"
echo " Date:     $(date -u)"
echo "============================================================"

TOTAL_TESTS=25
PASSED_TESTS=0
FAILED=0

pass_test() {
  echo "  â   RESULT: CONTAINED / ENFORCED ($1)"
  PASSED_TESTS=$((PASSED_TESTS + 1))
}

fail_test() {
  echo "  â   RESULT: POTENTIAL BREACH / UNEXPECTED BEHAVIOR ($1)" >&2
  FAILED=1
}

# --------------------------------------------------------------------------------
# SECTION 0: POSITIVE BASELINE CONTROL (SANITY CHECK)
# --------------------------------------------------------------------------------
echo ""
echo "â  â   [Baseline Control] Valid Execution Sanity Check â  â  "
```

```
BASELINE_CODE='print("BASELINE_VALID_EXECUTION")'
BASELINE_PAYLOAD=$(jq -n \
  --arg code "$BASELINE_CODE" \
  '{source_code: $code, language_id: 71, cpu_time_limit: 2.0, wall_time_limit: 4
> .0, memory_limit: 128000, enable_network: false}')

BASELINE_RESP=$(curl -s -X POST "${HEADERS[@]}" -d "$BASELINE_PAYLOAD" "$ENDPOIN
> T/submissions?wait=true&base64_encoded=false" || echo "")

if [ -z "$BASELINE_RESP" ]; then
  echo "[FATAL] Runner failed to respond to baseline valid submission. Aborting
> suite." >&2
  exit 1
fi

BASE_STATUS=$(echo "$BASELINE_RESP" | jq -r '.status.id // 0')
BASE_STDOUT=$(echo "$BASELINE_RESP" | jq -r '.stdout // ""' | tr -d '\r\n')

if [ "$BASE_STATUS" -ne 3 ] || [ "$BASE_STDOUT" != "BASELINE_VALID_EXECUTION" ];
>  then
  echo "[FATAL] Positive control failed! Status: $BASE_STATUS, Stdout: '$BASE_ST
> DOUT'. Response: $BASELINE_RESP" >&2
  echo "[FATAL] The runner cannot reliably execute valid code. Containment tests
>  cannot be evaluated." >&2
  exit 1
fi
echo "  â   Baseline valid code executed successfully (Status 3: Accepted)."

# ------------------------------------------------------------------------------
# SECTION 1: AUTHENTICATION & ACCESS CONTROL TESTS
# ------------------------------------------------------------------------------
echo ""
echo "â  â   [Test 1/$TOTAL_TESTS] Authentication Guard (Missing Token) â  â  "
HTTP_NO_AUTH=$(curl -s -o /dev/null -w "%{http_code}" -X POST -H "Content-Type:
> application/json" -d '{"source_code":"print(1)","language_id":71}' "$ENDPOIN
> T/submissions?wait=true" || echo "000")
if [ "$HTTP_NO_AUTH" = "401" ]; then
  pass_test "HTTP 401 on missing X-Auth-Token"
else
  fail_test "Expected HTTP 401, got HTTP $HTTP_NO_AUTH"
fi

echo ""
echo "â  â   [Test 2/$TOTAL_TESTS] Authentication Guard (Invalid Token) â  â  "
HTTP_BAD_AUTH=$(curl -s -o /dev/null -w "%{http_code}" -X POST -H "Content-Type:
>  application/json" -H "X-Auth-Token: bad_invalid_token_999" -d '{"source_cod
> e":"print(1)","language_id":71}' "$ENDPOINT/submissions?wait=true" || echo "
> 000")
if [ "$HTTP_BAD_AUTH" = "401" ]; then
  pass_test "HTTP 401 on invalid X-Auth-Token"
else
  fail_test "Expected HTTP 401, got HTTP $HTTP_BAD_AUTH"
fi

echo ""
echo "â  â   [Test 3/$TOTAL_TESTS] Administrative Authorization Guard (/system_info)
>  â  â  "
HTTP_ADMIN_NO_AUTH=$(curl -s -o /dev/null -w "%{http_code}" -X GET -H "X-Auth-To
> ken: $AUTHN_TOKEN" "$ENDPOINT/system_info" || echo "000")
if [ "$HTTP_ADMIN_NO_AUTH" = "401" ] || [ "$HTTP_ADMIN_NO_AUTH" = "403" ]; then
  pass_test "Admin endpoint rejected without X-Auth-User"
else
```

```
  fail_test "Expected HTTP 401/403 on /system_info without admin token, got HTTP
>  $HTTP_ADMIN_NO_AUTH"
fi

# ------------------------------------------------------------------------------
# HELPER: CODE EXECUTION RUNNER
# ------------------------------------------------------------------------------
run_code_test() {
  local num="$1"
  local title="$2"
  local lang_id="$3"
  local code="$4"
  local expected_status="$5"
  local check_pattern="$6"
  local extra_payload="${7:-}"

  echo ""
  echo "â  â   [Test $num/$TOTAL_TESTS] $title â  â  "

  local payload
  if [ -n "$extra_payload" ]; then
    payload=$(jq -n \
      --arg code "$code" \
      --argjson lang "$lang_id" \
      --argjson extra "$extra_payload" \
      '{source_code: $code, language_id: $lang, cpu_time_limit: 2.0, wall_time_l
> imit: 4.0, memory_limit: 128000} + $extra')
  else
    payload=$(jq -n \
      --arg code "$code" \
      --argjson lang "$lang_id" \
      '{source_code: $code, language_id: $lang, cpu_time_limit: 2.0, wall_time_l
> imit: 4.0, memory_limit: 128000, enable_network: false}')
  fi

  local response
  response=$(curl -s -X POST "${HEADERS[@]}" -d "$payload" "$ENDPOINT/submission
> s?wait=true&base64_encoded=false" || echo "")

  if [ -z "$response" ]; then
    fail_test "Curl execution failure: empty response from runner"
    return
  fi

  local status_id
  status_id=$(echo "$response" | jq -r '.status.id // 0' 2>/dev/null || echo 0)
  local status_desc
  status_desc=$(echo "$response" | jq -r '.status.description // "Unknown"' 2>/d
> ev/null || echo "Unknown")
  local stderr
  stderr=$(echo "$response" | jq -r '.stderr // ""' 2>/dev/null || echo "")
  local stdout
  stdout=$(echo "$response" | jq -r '.stdout // ""' 2>/dev/null || echo "")
  local compile_out
  compile_out=$(echo "$response" | jq -r '.compile_output // ""' 2>/dev/null ||
> echo "")

  echo "  Status ID:   $status_id ($status_desc)"
  if [ -n "$stderr" ]; then echo "  Stderr:      $(echo "$stderr" | head -n 2)";
>  fi
  if [ -n "$stdout" ]; then echo "  Stdout:      $(echo "$stdout" | head -n 1 |
> cut -c 1-80)"; fi
```

```
147 |
148 |   # Reject Status 13 (Internal Error) - an internal server crash is NEVER securi
    > ty containment
149 |   if [ "$status_id" -eq 13 ]; then
150 |     fail_test "Runner returned Status 13 (Internal Error) - infrastructure error
    > , not security containment"
151 |     return
152 |   fi
153 |
154 |   local matched=false
155 |   # Split expected_status by '|' and check exact match
156 |   IFS='|' read -ra ALLOWED_STATUSES <<< "$expected_status"
157 |   for s in "${ALLOWED_STATUSES[@]}"; do
158 |     if [ "$status_id" -eq "$s" ]; then
159 |       if [ -n "$check_pattern" ]; then
160 |         if echo "$stderr$stdout$compile_out$status_desc" | grep -qiE "$check_pat
    > tern"; then
161 |           matched=true
162 |           break
163 |         fi
164 |       else
165 |         matched=true
166 |         break
167 |       fi
168 |     fi
169 |   done
170 |
171 |   if [ "$matched" = true ]; then
172 |     pass_test "Defended by isolate/cgroups"
173 |   else
174 |     fail_test "Expected: Status in [$expected_status] matching '$check_pattern',
    >  Got: Status $status_id ($status_desc)"
175 |   fi
176 | }
177 |
178 | # ------------------------------------------------------------------------------
179 | # SECTION 2: NETWORK ISOLATION & OVERRIDE ATTEMPTS
180 | # ------------------------------------------------------------------------------
181 | CODE_NET='
182 | import socket
183 | s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
184 | s.settimeout(1.0)
185 | try:
186 |     s.connect(("8.8.8.8", 53))
187 |     print("VULNERABLE_NETWORK_OPEN")
188 | except Exception as e:
189 |     print("NETWORK_BLOCKED:", type(e).__name__)
190 | '
191 | run_code_test 4 "Outbound Network Egress (Default)" 71 "$CODE_NET" "3" "NETWORK_
    > BLOCKED"
192 |
193 | # Test 5: Client-side network override attempt (enable_network=true)
194 | # Must still be blocked because ALLOW_ENABLE_NETWORK=false in judge0.conf
195 | run_code_test 5 "Client Network Override (enable_network=true)" 71 "$CODE_NET" "
    > 3|11|12" "NETWORK_BLOCKED|Operation not permitted|Network is unreachable|Can
    > not enable network" '{"enable_network": true}'
196 |
197 | CODE_META='
198 | import urllib.request
199 | try:
200 |     res = urllib.request.urlopen("http://169.254.169.254/latest/meta-data/", tim
    > eout=1.0)
```

```
    print("VULNERABLE_METADATA_ACCESSED")
except Exception as e:
    print("METADATA_BLOCKED:", type(e).__name__)
'
run_code_test 6 "Cloud Metadata IP Access (169.254.169.254)" 71 "$CODE_META" "3"
>  "METADATA_BLOCKED"

# ------------------------------------------------------------------------------
# SECTION 3: RESOURCE CONSTRAINTS & MULTI-PROCESS CONTAINMENT
# ------------------------------------------------------------------------------
CODE_FORK='
import os, time
count = 0
try:
    while True:
        pid = os.fork()
        if pid == 0:
            time.sleep(10)
            os._exit(0)
        count += 1
except Exception as e:
    print(f"FORK_BOMB_CONTAINED: at {count} forks ({type(e).__name__})")
'
run_code_test 7 "Process Fork Bomb" 71 "$CODE_FORK" "3|11|12" "FORK_BOMB_CONTAIN
> ED|BlockingIOError|Resource temporarily unavailable"

# Single process memory bomb
CODE_MEM_SINGLE='
chunks = []
try:
    for i in range(100):
        chunks.append(b"X" * (10 * 1024 * 1024))
    print("VULNERABLE_MEMORY_LEAK")
except MemoryError:
    print("MEMORY_LIMIT_CONTAINED: MemoryError")
'
run_code_test 8 "Single-Process Heap Memory Exhaustion" 71 "$CODE_MEM_SINGLE" "3
> |11|12" "MEMORY_LIMIT_CONTAINED|Memory limit exceeded|MemoryError"

# Multi-process memory bomb (Verifies aggregate cgroup memory across child proce
> sses)
CODE_MEM_MULTI='
import os, sys, time
# Fork 5 children each allocating 30MB (Total = 150MB > 128MB limit)
for i in range(5):
    pid = os.fork()
    if pid == 0:
        b = b"Y" * (30 * 1024 * 1024)
        time.sleep(2)
        os._exit(0)
for i in range(5):
    os.wait()
print("VULNERABLE_MULTI_PROCESS_MEMORY_LEAK")
'
run_code_test 9 "Multi-Process Aggregate Cgroup Memory Bomb" 71 "$CODE_MEM_MULTI
> " "11|12" "Memory limit exceeded|Killed|NZEC"

CODE_CPU='
# Infinite loop consuming 100% CPU
i = 0
while True:
    i += 1
```

```
258 | ’
259 | run_code_test 10 "Infinite CPU Loop (CPU & Wall Timeout)" 71 "$CODE_CPU" "5" "Ti
    > me Limit Exceeded"
260 |
261 | CODE_DISK=’
262 | try:
263 |     with open("bigfile.tmp", "wb") as f:
264 |         f.write(b"0" * (25 * 1024 * 1024))
265 |     print("VULNERABLE_DISK_FLOOD")
266 | except Exception as e:
267 |     print("DISK_LIMIT_CONTAINED:", type(e).__name__)
268 | ’
269 | run_code_test 11 "Disk Space Exhaustion (/box quota)" 71 "$CODE_DISK" "3|8|11" "
    > DISK_LIMIT_CONTAINED|SIGXFSZ|File size limit exceeded"
270 |
271 | # ------------------------------------------------------------------------
272 | # SECTION 4: PRIVILEGE ESCALATION & HOST INTEGRITY
273 | # ------------------------------------------------------------------------
274 | CODE_PASSWD=’
275 | try:
276 |     with open("/etc/passwd", "a") as f:
277 |         f.write("hacker:x:0:0::/:/bin/sh\n")
278 |     print("VULNERABLE_ROOT_OVERWRITE")
279 | except Exception as e:
280 |     print("PASSWD_READONLY_CONTAINED:", type(e).__name__)
281 | ’
282 | run_code_test 12 "Host Filesystem Overwrite (/etc/passwd)" 71 "$CODE_PASSWD" "3"
    >  "PASSWD_READONLY_CONTAINED"
283 |
284 | CODE_ENV=’
285 | import os
286 | secrets = [v for k, v in os.environ.items() if "AUTH" in k or "PASS" in k or "SE
    > CRET" in k or "TOKEN" in k]
287 | if secrets:
288 |     print("VULNERABLE_SECRETS_FOUND")
289 | else:
290 |     print("ENV_SECRETS_SANITIZED")
291 | ’
292 | run_code_test 13 "Environment Secret Sniffing" 71 "$CODE_ENV" "3" "ENV_SECRETS_S
    > ANITIZED"
293 |
294 | CODE_PROC=’
295 | import os
296 | try:
297 |     pids = [int(p) for p in os.listdir("/proc") if p.isdigit()]
298 |     # Isolate PID namespace hides all host/container processes except sandbox in
    > it & self
299 |     if len(pids) <= 3:
300 |         print(f"PID_NAMESPACE_ISOLATED: {len(pids)} PIDs visible")
301 |     else:
302 |         print(f"VULNERABLE_HOST_PIDS_VISIBLE: {len(pids)} PIDs")
303 | except Exception as e:
304 |     print("PROC_ACCESS_DENIED:", type(e).__name__)
305 | ’
306 | run_code_test 14 "Host Process Sniffing (/proc)" 71 "$CODE_PROC" "3" "PID_NAMESP
    > ACE_ISOLATED|PROC_ACCESS_DENIED"
307 |
308 | CODE_PRIV=’
309 | import ctypes
310 | try:
311 |     libc = ctypes.CDLL("libc.so.6")
312 |     res = libc.setuid(0)
```

```
    if res == 0:
        print("VULNERABLE_SETUID_ROOT_SUCCESS")
    else:
        print("PRIVILEGE_ESCALATION_BLOCKED")
except Exception as e:
    print("PRIVILEGE_ESCALATION_BLOCKED:", type(e).__name__)
'
run_code_test 15 "Privilege Escalation (setuid 0)" 71 "$CODE_PRIV" "3|11" "PRIVI
> LEGE_ESCALATION_BLOCKED"

CODE_RAW_SOCK='
import socket
try:
    s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_ICMP)
    print("VULNERABLE_RAW_SOCKET_OPENED")
except Exception as e:
    print("RAW_SOCKET_BLOCKED:", type(e).__name__)
'
run_code_test 16 "Raw Socket Packet Sniffing (CAP_NET_RAW)" 71 "$CODE_RAW_SOCK"
> "3|11" "RAW_SOCKET_BLOCKED|PermissionError"

# ------------------------------------------------------------------------------
# SECTION 5: MULTI-LANGUAGE SANDBOX CONTAINMENT (C++, Java, JavaScript)
# ------------------------------------------------------------------------------

# C++ (Language ID 54)
CODE_CPP_CPU='
#include <iostream>
int main() {
    while (true) {}
    return 0;
}
'
run_code_test 17 "C++ Infinite CPU Loop (Timeout Enforcement)" 54 "$CODE_CPP_CPU
> " "5" "Time Limit Exceeded"

CODE_CPP_MEM='
#include <iostream>
#include <vector>
int main() {
    try {
        std::vector<char*> chunks;
        for (int i = 0; i < 50; ++i) {
            chunks.push_back(new char[20 * 1024 * 1024]);
        }
        std::cout << "VULNERABLE_CPP_MEMORY_LEAK" << std::endl;
    } catch (...) {
        std::cout << "CPP_MEMORY_CONTAINED" << std::endl;
    }
    return 0;
}
'
run_code_test 18 "C++ Heap Memory Exhaustion" 54 "$CODE_CPP_MEM" "3|11|12" "CPP_
> MEMORY_CONTAINED|Memory limit exceeded|bad_alloc|Killed"

CODE_CPP_FORK='
#include <unistd.h>
#include <iostream>
int main() {
    int count = 0;
    while (true) {
        pid_t p = fork();
```

```
        if (p < 0) {
            std::cout << "CPP_FORK_BOMB_CONTAINED" << std::endl;
            break;
        }
        if (p == 0) {
            pause();
            return 0;
        }
        count++;
    }
    return 0;
}
'
run_code_test 19 "C++ Excessive Process Spawning (Fork Bomb)" 54 "$CODE_CPP_FORK
> " "3|11|12" "CPP_FORK_BOMB_CONTAINED|Resource temporarily unavailable|Blocki
> ngIOError"

CODE_CPP_NET='
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <iostream>
int main() {
    int s = socket(AF_INET, SOCK_STREAM, 0);
    struct sockaddr_in serv;
    serv.sin_family = AF_INET;
    serv.sin_port = htons(53);
    inet_pton(AF_INET, "8.8.8.8", &serv.sin_addr);
    if (connect(s, (struct sockaddr*)&serv, sizeof(serv)) < 0) {
        std::cout << "CPP_NETWORK_BLOCKED" << std::endl;
    } else {
        std::cout << "VULNERABLE_CPP_NETWORK_OPEN" << std::endl;
    }
    close(s);
    return 0;
}
'
run_code_test 20 "C++ Outbound Socket Connection" 54 "$CODE_CPP_NET" "3" "CPP_NE
> TWORK_BLOCKED"

# Java (Language ID 62)
CODE_JAVA_CPU='
public class Main {
    public static void main(String[] args) {
        while (true) {}
    }
}
'
run_code_test 21 "Java Infinite CPU Loop (Timeout Enforcement)" 62 "$CODE_JAVA_C
> PU" "5" "Time Limit Exceeded"

CODE_JAVA_MEM='
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        try {
            ArrayList<byte[]> list = new ArrayList<>();
            while (true) {
                list.add(new byte[10 * 1024 * 1024]);
            }
        } catch (OutOfMemoryError e) {
```

```
            System.out.println("JAVA_MEMORY_CONTAINED");
        }
    }
}
'
run_code_test 22 "Java Heap Memory Exhaustion" 62 "$CODE_JAVA_MEM" "3|11|12" "JA
> VA_MEMORY_CONTAINED|OutOfMemoryError|Memory limit exceeded|Killed"

# JavaScript (Language ID 63)
CODE_JS_CPU='
while (true) {}
'
run_code_test 23 "JavaScript Infinite CPU Loop (Timeout Enforcement)" 63 "$CODE_
> JS_CPU" "5" "Time Limit Exceeded"

CODE_JS_MEM='
const chunks = [];
try {
    while (true) {
        chunks.push(Buffer.alloc(10 * 1024 * 1024));
    }
} catch (e) {
    console.log("JS_MEMORY_CONTAINED");
}
'
run_code_test 24 "JavaScript Heap Memory Exhaustion" 63 "$CODE_JS_MEM" "3|11|12"
>  "JS_MEMORY_CONTAINED|allocation failure|out of memory|Memory limit exceeded
> "

CODE_JS_NET='
const net = require("net");
const client = net.createConnection({ port: 80, host: "1.1.1.1" }, () => {
    console.log("VULNERABLE_JS_NETWORK_OPEN");
    client.end();
});
client.on("error", (err) => {
    console.log("JS_NETWORK_BLOCKED: " + err.code);
});
setTimeout(() => {}, 500);
'
run_code_test 25 "JavaScript Outbound Network Connection" 63 "$CODE_JS_NET" "3"
> "JS_NETWORK_BLOCKED"

echo ""
echo "========================================================="
echo " Verification Complete: $PASSED_TESTS / $TOTAL_TESTS Tests Contained"
echo "========================================================="

if [ "$PASSED_TESTS" -eq "$TOTAL_TESTS" ] && [ "$FAILED" -eq 0 ]; then
  echo " STATUS: ALL HOSTILE ATTACK VECTORS CONTAINED."
  exit 0
else
  echo " STATUS: POTENTIAL VULNERABILITY DETECTED. DO NOT PROMOTE TO PRODUCTION.
> "
  exit 1
fi

```

5. Hardened Docker Compose Stack Configuration

File Path: deploy/judge0/docker-compose.yml [COMPOSE SPECIFICATION (YAML)]

Production container stack configuration: fail-closed ARM64 image requirement, 4992 MB aggregate memory limits, and isolated internal bridge.

```
 1 | version: '3.8'
 2 |
 3 | # ==============================================================================
 4 | # TestForge Dedicated Execution Runner - Docker Compose Stack
 5 | # Pinned Image: Judge0 CE v1.13.1
 6 | # Security Notice: Privileged mode is strictly confined to judge0-server and
 7 | # judge0-workers as required by isolate. The Docker socket is NEVER mounted.
 8 | # Host cgroups are managed dynamically by the container engine.
 9 | # ==============================================================================
10 |
11 | services:
12 |   judge0-server:
13 |     image: ${JUDGE0_IMAGE:?FATAL: JUDGE0_IMAGE must be set in .env (testforge/ju
   > dge0:v1.13.1-arm64 on ARM64)}
14 |     container_name: testforge-judge0-server
15 |     restart: always
16 |     privileged: false # Server strictly handles HTTP Puma routing and does not c
   > all isolate
17 |     mem_limit: 1024m
18 |     cpus: 0.50
19 |     env_file:
20 |       - judge0.conf
21 |     environment:
22 |       - REDIS_HOST=judge0-redis
23 |       - REDIS_PORT=6379
24 |       - REDIS_PASSWORD=${REDIS_PASSWORD:-judge0_secret_redis_pass}
25 |       - POSTGRES_HOST=judge0-db
26 |       - POSTGRES_USER=${POSTGRES_USER:-judge0}
27 |       - POSTGRES_PASSWORD=${POSTGRES_PASSWORD:-judge0_secret_db_pass}
28 |       - POSTGRES_DB=${POSTGRES_DB:-judge0}
29 |     ports:
30 |       # Bind exclusively to 127.0.0.1 so it cannot be reached from the public in
   > ternet without TLS reverse proxy
31 |       - '127.0.0.1:2358:2358'
32 |     depends_on:
33 |       judge0-db:
34 |         condition: service_healthy
35 |       judge0-redis:
36 |         condition: service_healthy
37 |     networks:
38 |       - judge0-net
39 |
40 |   judge0-workers:
41 |     image: ${JUDGE0_IMAGE:?FATAL: JUDGE0_IMAGE must be set in .env (testforge/ju
   > dge0:v1.13.1-arm64 on ARM64)}
42 |     container_name: testforge-judge0-workers
43 |     command: ["./scripts/workers"]
44 |     restart: always
```

```
45 |     privileged: true # Required for isolate to manage mount/net namespaces, cgro
   > up v1, and chroot
46 |     mem_limit: 3072m
47 |     cpus: 1.00
48 |     env_file:
49 |       - judge0.conf
50 |     environment:
51 |       - COUNT=1 # Staging safety policy: single concurrent execution until runti
   > me measurements justify scaling
52 |       - REDIS_HOST=judge0-redis
53 |       - REDIS_PORT=6379
54 |       - REDIS_PASSWORD=${REDIS_PASSWORD:-judge0_secret_redis_pass}
55 |       - POSTGRES_HOST=judge0-db
56 |       - POSTGRES_USER=${POSTGRES_USER:-judge0}
57 |       - POSTGRES_PASSWORD=${POSTGRES_PASSWORD:-judge0_secret_db_pass}
58 |       - POSTGRES_DB=${POSTGRES_DB:-judge0}
59 |     depends_on:
60 |       judge0-db:
61 |         condition: service_healthy
62 |       judge0-redis:
63 |         condition: service_healthy
64 |     networks:
65 |       - judge0-net
66 |     # Note: Host /sys/fs/cgroup is NOT mounted read-only (:ro).
67 |     # In privileged mode, Docker provides read-write cgroups directly into the c
   > ontainer.
68 |
69 |   judge0-db:
70 |     image: postgres:16-alpine
71 |     container_name: testforge-judge0-db
72 |     restart: always
73 |     mem_limit: 512m
74 |     cpus: 0.25
75 |     environment:
76 |       - POSTGRES_USER=${POSTGRES_USER:-judge0}
77 |       - POSTGRES_PASSWORD=${POSTGRES_PASSWORD:-judge0_secret_db_pass}
78 |       - POSTGRES_DB=${POSTGRES_DB:-judge0}
79 |     volumes:
80 |       - judge0_db_data:/var/lib/postgresql/data
81 |     networks:
82 |       - judge0-net
83 |     healthcheck:
84 |       test: ['CMD-SHELL', 'pg_isready -U judge0']
85 |       interval: 10s
86 |       timeout: 5s
87 |       retries: 5
88 |
89 |   judge0-redis:
90 |     image: redis:7-alpine
91 |     container_name: testforge-judge0-redis
92 |     restart: always
93 |     mem_limit: 384m
94 |     cpus: 0.25
95 |     command: [
96 |       'redis-server',
97 |       '--appendonly', 'yes',
98 |       '--requirepass', '${REDIS_PASSWORD:-judge0_secret_redis_pass}',
99 |       '--maxmemory', '256mb',
100 |       '--maxmemory-policy', 'allkeys-lru'
101 |     ]
102 |     volumes:
103 |       - judge0_redis_data:/data
```

```
104 |     networks:
105 |       - judge0-net
106 |     healthcheck:
107 |       test: [’CMD’, ’redis-cli’, ’-a’, ’${REDIS_PASSWORD:-judge0_secret_redis_pa
    > ss}’, ’ping’]
108 |       interval: 10s
109 |       timeout: 5s
110 |       retries: 5
111 |
112 | networks:
113 |   judge0-net:
114 |     driver: bridge
115 |     internal: false # Requires internet for apt/package initialization, isolate
    > disables egress per box
116 |
117 | volumes:
118 |   judge0_db_data:
119 |   judge0_redis_data:
120 |
```

6. Custom Native ARM64 Judge0 v1.13.1 Dockerfile

File Path: deploy/judge0/docker/Dockerfile.judge0-arm64 [DOCKERFILE]

Native ARM64 Judge0 worker & server image: clean source build, scoped sudoers (0440), and root-owned start-cron-daemon wrapper.

```
# ================================================================================
# TestForge Dedicated Execution Runner - Custom Native ARM64 Judge0 v1.13.1 Image
# Target Platform: linux/arm64 (OCI Ampere A1, AWS Graviton, Apple Silicon)
# Base Image: testforge/compilers:v1.4.0-arm64
# Preserves: Judge0 CE v1.13.1 behavior, Rails 6 API, ActiveJob worker, and Isolate
# ================================================================================

FROM testforge/compilers:v1.4.0-arm64 AS production

ENV JUDGE0_HOMEPAGE="https://judge0.com"
LABEL homepage=$JUDGE0_HOMEPAGE

ENV JUDGE0_SOURCE_CODE="https://github.com/judge0/judge0"
LABEL source_code=$JUDGE0_SOURCE_CODE

ENV JUDGE0_MAINTAINER="TestForge DevSecOps Team <security@testforge.internal>"
LABEL maintainer=$JUDGE0_MAINTAINER

ENV PATH="/usr/local/ruby-2.7.0/bin:/opt/.gem/bin:$PATH"
ENV GEM_HOME="/opt/.gem/"
ENV RAILS_ENV=production

# 1. Environment & Directories
# Note: Aglio documentation generator is omitted to prevent build failures on Node 20 / ARM64 (B-07).
EXPOSE 2358
WORKDIR /api

# 2. Acquire Judge0 CE v1.13.1 Source Code at Verified Release Tag
# Copy complete source tree including dotfiles cleanly without error suppression (B-07)
RUN git clone --depth 1 --branch v1.13.1 https://github.com/judge0/judge0.git /tmp/judge0-src && \
    cp -a /tmp/judge0-src/. /api/ && \
    rm -rf /tmp/judge0-src

# 3. Install Ruby Gems for Rails 6.0 in Production
RUN bundle config set --local deployment 'true' && \
    bundle config set --local without 'development test' && \
    bundle install --jobs=$(nproc)

# 4. Configure & Validate System Cron for Periodic Maintenance
# System crontabs in /etc/cron.d must have 0644 root:root permissions and a trailing newline.
# They MUST NOT be piped into a user crontab because system crontabs contain an explicit user field.
```

```
RUN if [ -d "/api/cron" ] && [ -n "$(ls -A /api/cron 2>/dev/null)" ]; then \
      cp /api/cron/* /etc/cron.d/ 2>/dev/null || true; \
      chown -R root:root /etc/cron.d/ && \
      chmod 0644 /etc/cron.d/* && \
      for f in /etc/cron.d/*; do \
        [ -f "$f" ] || continue; \
        sed -i -e '$a\' "$f"; \
        echo "Validated system crontab format for $f"; \
      done; \
    fi

# 5. Create Root-Owned Service Wrapper & Scoped Sudoers Configuration
# Creates a dedicated wrapper to prevent granting broad service management permissions.
RUN printf '#!/bin/sh\nset -eu\nif command -v service >/dev/null 2>&1; then\n  service cron start\nelse\n  /etc/init.d/cron start\nfi\n' > /usr/local/bin/start-cron-daemon && \
    chmod 0755 /usr/local/bin/start-cron-daemon && \
    chown root:root /usr/local/bin/start-cron-daemon && \
    if ! id -u judge0 >/dev/null 2>&1; then \
      useradd -u 1000 -m -r judge0; \
    fi && \
    echo "judge0 ALL=(ALL) NOPASSWD: /usr/local/bin/start-cron-daemon, /usr/local/bin/isolate, /usr/sbin/service cron start, /etc/init.d/cron start" > /etc/sudoers.d/judge0 && \
    chmod 0440 /etc/sudoers.d/judge0 && \
    mkdir -p /api/tmp/pids /api/tmp/sockets /api/log && \
    chown -R judge0:judge0 /api/tmp/ /api/log/

USER judge0

ENV JUDGE0_VERSION="1.13.1-arm64"
LABEL version=$JUDGE0_VERSION

ENTRYPOINT ["/api/docker-entrypoint.sh"]
CMD ["/api/scripts/server"]

# Development stage for interactive debugging
FROM production AS development
CMD ["sleep", "infinity"]

```